

Quentin Adolphe

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science | Edge AI specialization

Expected 2029

Swarthmore College

Bachelor of Arts in Computer Science, Minor in Engineering | GPA: 3.9/4.0

Class of 2025

PROFESSIONAL EXPERIENCE

General Motors

Robotics AI/ML Software Engineer

Sep 2026 - Present

- Integrate and test motion-planning and scene-perception software for autonomous systems.

General Motors

Dimensional Systems Engineer

Mar 2026 - Sep 2026

- Replaced a manual Excel/VBA workflow with a Databricks ETL pipeline, processing approximately 100,000 deliverable rows across vehicle programs to improve DSE visibility and execution.
- Developed a 3D variational autoencoder tool to edit tolerance models, run simulations, analyze results, and generate reports, saving more than 40 hours per simulation and reporting cycle.

General Motors

CFD Automation Engineer

Jul 2025 - Mar 2026

- Built a Python analytics framework to evaluate tire parameters affecting virtual vehicle stability.
- Rebuilt a legacy Tkinter GUI as a React, Electron, and Flask application to streamline preprocessing, HPC submission, monitoring, and postprocessing work.

Swarthmore College

Teaching Assistant - Computer Vision & Mobile Robotics

Jan 2024 - May 2025

- Helped students work through labs in perception algorithms, robot kinematics, and control systems.
- Reviewed and debugged assignments for 20+ students, giving feedback on code and problem-solving approaches.

General Motors

Software Validation Engineer Intern

May 2024 - Aug 2024

- Built applications supporting flash development, including a custom utility file generator.
- Automated ECU flash validation against Software-Defined Vehicle specifications.

USAA

Software Engineering Intern

May 2023 - Aug 2023

- Contributed to full-stack development of a modernized life insurance application with database integration.
- Implemented pages with React and JavaScript and created application metadata.

PROJECTS

Running Analysis with Machine Learning | Kotlin, TensorFlow, Computer Vision, Machine Learning

- Used TensorFlow's Thunder convolutional neural network to estimate a sprinter's joint positions.
- Trained a neural network to identify the current phase of a sprinter in motion.

3D Triangulation of Objects | Python, Computer Vision, Stereo Calibration

- Calibrated a stereo camera setup to calculate 3D real-world coordinates from 2D images.

Turtlebot Maze Traversal | Python, Robotics, Dijkstra's Algorithm, Control Systems

- Implemented Dijkstra's algorithm and a proportional-derivative controller to navigate a robot through a maze.

SKILLS

Programming: Python, Java, JavaScript, TypeScript

Frameworks and Platforms: React, Node.js, Flask, Databricks

AI and Robotics: Machine Learning, Computer Vision, Robotics

Data and Automation: Data Aggregation, Process Automation, ETL Pipelines, MCP Servers